MATH 526 Section 001 Lab 6 Fall 2004

Part I

We learn to plot 2D graphs in this part. First issue the command

>>**help plot**

to get the basic information on **plot**.

We can draw the graph of the sine function over the interval $[-4, 4]$ using a partition of mesh size 0.01 as follows:

>>**x = –4: .01: 4;**

This creates the vector $\mathbf{x} = [-4\ \ -3.99\ \ -3.98\ \cdots\ 3.98\ 3.99\ 4]$ with 801 components.

>>**y = sin(x);**

This creates the vector **y** whose components are the sines of the corresponding components of **x**.

>>**plot(x,y)**

The graph of the function $y = \sin x$ over the interval $[-4, 4]$ is displayed in a figure window.

To clear the figure window, type

>>**clf**

We now draw the graph of $y = e^{x^2}$ over the interval $[-1.5, 1.5]$ using a partition of mesh size 0.01.

>>**clear x y**

>>**x = –1.5: .01: 1.5;**

This creates the vector $\mathbf{x} = [-1.5\ \ -1.49\ \ -1.48\ \cdots\ 1.48\ 1.49\ 1.5]$ with 301 components.

>>**y = exp(x.^2);**

x.^**2** is the vector whose components are the squares of the corresponding components of **x**. (Use **help arith** to learn more about array multiplication and array power.) The command **exp**(**x**.^**2**) then creates the vector **y** whose components are the exponentials of the corresponding components of the vector **x**.^**2**. The net effect is that the components of **y** are the exponentials of the squares of the corresponding components of **x**.

>>**plot(x,y)**

The graph of the function $y = \exp(x^2)$ over the interval $[-1.5, 1.5]$ is now displayed in the figure window.

We can add labels and title to the figure as follows:

>>**xlabel('x-axis')**

>>**ylabel('y-axis')**

>>**title('My second plot')**

Of course you can also check out **help xlabel**, **help ylabel** and **help title**.

We can also change the fill of the line used. For example, see what the following commands do:

>>**clf**
>>**plot(x,y,'–.')**

Create two other vectors **s** and **t** by the following commands:

>>**t=–1.5:.1:1.5**

The vector **t** contains the points that divide the interval $[-1.5, 1.5]$ into subintervals of length 0.1.

>>**s=t.^3;**

The components of the vector **s** are the cubes of the corresponding components of **t**.

Now plot the graph of the cubic polynomial $s = t^3$ over the interval $[-1.5, 1.5]$ by the command

>>**plot(t,s)**

The graph of $y = \exp(x^2)$ in the figure window is now replaced by the graph of $s = t^3$.

We can add the graph of $y = \exp(x^2)$ back to the figure window by typing the following commands.

>>**hold on**

The graph of $s = t^3$ is now held in the figure window.

>>**plot(x,y)**

Both graphs now appear in the figure window. Use **help hold** to learn more about the command **hold**.

Another way of putting two (or more) graphs in the same figure is to plot them simultaneously. Type the following commands:

>>**figure**

A new figure window is created.

>>**plot(x,exp(x.^2),'r',x,x.^3,'b')**

The graph of $y = \exp(x^2)$ now appears as a red curve in the new figure window and the graph of $y = x^3$ appears as a blue curve.

To close the figure windows, type

>>**close all**

Clear all variables before you work on Part II.